Bhuvanya Sirigineedi

ML Engineer / Software Engineer / Data Scientist

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PROFESSIONAL SUMMARY:

- Machine Learning Engineer with 5+ years of experience in developing and deploying scalable Machine Learning and Deep Learning models using Python, PyTorch, TensorFlow, Scikit-learn, and Keras. Proficient in end-to-end pipelines including Feature Engineering, Predictive Modeling, NLP, Computer Vision, Neural Networks, Model Deployment, MLOps, Docker, Kubernetes, CI/CD, Model Monitoring, and Pipeline Automation on AWS, Azure, and GCP. Skilled in Statistics, Data Modeling, Data Mining, and cross-functional collaboration to deliver production-grade solutions for predictive maintenance, anomaly detection, and optimization in energy and analytics environments. Strong problem-solving and communication abilities for driving technical and business impact.

SKILLS:

- **Programming & Languages:** Python, Java, SQL, Algorithms, Data Structures
- **ML/DL Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face
- **Machine Learning & AI Domains:** Machine Learning, Deep Learning, NLP, Computer Vision, Neural Networks, Reinforcement Learning, Predictive Modeling, Generative AI, LLM, Prompt Engineering, RAG, Transformers, RLHF
- **Cloud & Infrastructure:** AWS, Azure, GCP, Scalable Systems, Model Deployment
- **MLOps & DevOps:** MLOps, Docker, Kubernetes, CI/CD, Model Monitoring, Pipeline Automation
- **Data & Analytics:** Statistics, Data Modeling, Data Mining, Feature Engineering, Tableau, Power BI
- **Professional:** Communication, Cross-functional Collaboration, Problem Solving

PROFESSIONAL EXPERIENCE:

Chevron, TX

Software Engineer - ML, Jan 2025 to Present

- Developed and deployed scalable Machine Learning models for predictive maintenance using Python, PyTorch, TensorFlow, and Scikit-learn, incorporating Feature Engineering, Neural Networks, and time-series Predictive Modeling to support equipment health monitoring.
- Built end-to-end ML pipelines with MLOps practices, Docker, Kubernetes, CI/CD, and Model Monitoring on AWS, enabling automated Pipeline Automation and reliable Model Deployment in production, scalable systems.
- Applied Deep Learning and NLP techniques for anomaly detection and data analysis, collaborating cross-functionally with engineering and operations teams to integrate solutions and improve operational efficiency.
- Optimized models for performance using Statistics, Data Modeling, and problem-solving to ensure low-latency inference and high reliability in energy domain applications.

Fractal Analytics, India

Machine Learning Engineer, May 2021 to Jul 2023

- Designed production-grade Machine Learning solutions using Python, SQL, TensorFlow, Keras, Hugging Face, and Scikit-learn for predictive modeling, recommendation systems, and optimization across client projects.
- Implemented Deep Learning models with Transformers and Neural Networks for NLP and Computer Vision tasks, focusing on Feature Engineering, Data Mining, and model evaluation to deliver measurable improvements.
- Deployed models via MLOps workflows with Docker, Kubernetes, CI/CD, and Model Monitoring on cloud platforms (AWS, Azure, GCP), supporting scalable systems and Pipeline Automation.
- Conducted experiments with Reinforcement Learning elements and Generative AI approaches (including LLM and RAG techniques) to enhance model accuracy and business outcomes through cross-functional collaboration.
- Mentored team members on best practices in algorithms, data structures, statistics, and problem solving for robust ML development.

Fractal Analytics, India
Data Scientist, Jan 2021 to Apr 2021

- Built Machine Learning models using Python, SQL, Scikit-learn, and Keras for classification, regression, clustering, and NLP applications, emphasizing Feature Engineering and Data Modeling.
- Performed exploratory analysis, Data Mining, and Statistics-driven insights with tools like Tableau and Power BI to support stakeholder decision-making.
- Collaborated cross-functionally to translate requirements into technical solutions, applying predictive modeling and problem-solving to generate actionable results.

Tech Mahindra, India
Data Analyst Intern, Jun 2020 to Dec 2020

- Assisted in developing dashboards and data visualizations using tools like Excel, Power BI, or Tableau to present key performance indicators and business metrics to stakeholders.
- Collected, cleaned, and validated data from multiple sources to ensure accuracy and consistency for business analysis and reporting purposes
- Utilized SQL queries to extract and manipulate data from databases, enabling efficient data retrieval for various analytical projects.

EDUCATION:

University of Houston, Houston, TX
Master of Science in Data Science, May 2025
Sagi Ramakrishnam Raju Engineering College, India
Bachelor of Technology in Information Technologies, Aug 2021

CERTIFICATION:

AWS Machine Learning Engineer – Associate
Microsoft Certified Power BI Data Analyst Associate
AWS Certified Solutions Architect Associate

PROJECT:

Stock Market Analysis | Python, Feature Engineering, Yahoo Finance API Jan 2025 – Apr 2025
Implemented adversarial attack frameworks to evaluate robustness and semantic degradation in LLMs.
Designed evaluation metrics supporting safe and resilient AI system deployment.
Trained and tested deep learning models using PyTorch, aligning with modern ML research practices.
Adversarial Robustness in LLMs | PyTorch, LLM Models, Neural Networks, GIT Jan 2025 – Apr 2025
Built an end-to-end ETL and ML pipeline for financial time-series data using APIs.
Implemented feature engineering, trend analysis, and predictive modeling, producing visual dashboards.
Big Data Analytics on Economic Growth | Hadoop, MapReduce, Python Jan 2024 – Apr 2024
Processed 2M+ records using distributed computing frameworks.
Built regression models and optimized transformation logic for scalable analytics.
COVID-19 Vaccination Analysis on U.S.A | Python, EDA, ANOVA Aug 2023 – Dec 2023
Analyzed vaccination patterns across 1,000+ U.S. counties to identify demographic and socioeconomic predictors.
Built a time-series forecasting model to estimate when counties reach 80 percent + vaccination coverage, with insights shared to health science departments for outreach.